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Genome sequencing and annotation of multi-virulent *Aeromonas veronii* XhG1.2 isolated from diseased *Xiphophorus hellerii*

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Abstract

The present study was intended to elucidate the genomic basis of antibiotic resistance and hyper-virulence of the fish pathogen *Aeromonas veronii* XhG1.2 characterized in our previous work. The identity of XhG1.2 was confirmed through 16S rDNA sequence analysis and whole genome sequence analysis by illumina platform. The top-hit species distribution analysis of XhG1.2 revealed major hits against the *Aeromonas veronii*. The identification of virulence genes using the VFDB showed the genome of XhG1.2 to have the genes coding for the virulence factors viz. aerolysin, RtxA, T2SS, T3SS, T6SS. The presence of antibiotic resistance predicted through the CARD database analysis showed it to have the *Cph*A3, OXA-12, *aed*F and pulvomycin resistance genes. By the phylogenetic and comparative genomic analysis, *A. veronii* species were found to code for toxin genes and confirmed the

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pathogenicity and drug resistance of *A. veronii* XhG1.2 and also its potential to cause disease in diverse ornamental fishes.

Keywords: Whole genome sequencing; *Aeromonas veroniii* XhG1.2; Aerolysin; Haemolysin III; Antibiotic resistance

1. INTRODUCTION

Aeromonas mediated diseases of fresh water aquarium fishes generally result in huge economic loss. *Aeromonas veronii* of *Aeromonas* spp. has recently been identified to have association with several fish diseases. General method to manage the *Aeromonas* disease of ornamental fishes involves the application of antibiotics. However, farmers use the antibiotics unconditionally to resist the disease causing pathogens and thereby to reduce the mortality rate (Mo, Chen, Leung, & Leung, 2017). The use of antibiotics has also been reported to enhance the growth and body weight of fishes. Despite the instant advantages of such antibiotic applications, the practice is unscientific and have serious consequences as it significantly favour the evolution of antibiotic resistance (Kim et al., 2019; Limbu, Zhou, Sun, Zhang, & Du, 2018). As *Aeromonas* spp. are known to have distribution in wide range aquatic environments, the accumulated antibiotics in these habitats can also enhance the antibiotic resistance evolution. Even though the antibiotic sensitivity tests can provide a preliminary insight on this, whole genome based analysis is very important to get molecular confirmation of the same.

The group *Aeromonads* are known to cause diseases with similar symptoms such as haemorrhagic septicaemia and ulcerative syndrome. Although *A. hydrophila* is considered as a major fish pathogen, studies on the pathogenicity and virulence factor evolution among other species like *A. veronii* are also important (Janda & Abbott, 2010). Because, *A. veronii* has recently been identified to cause significant threat to the ornamental fishes. As the presence of *A. veronii* has also been reported from the environmental, clinical and food samples, its detailed genomic analysis is important (Ghenghesh et al., 2008). For the identification of challenges with such emerging *Aeromonas* spp., whole genome sequence (WGS) based analysis is very important. Genomic insight generated through WGS analysis can give accurate prediction of pathogenicity, virulence potential and antimicrobial resistance of *A. veronii*. With the advances in next generation sequencing (NGS) technologies, different sequencing platforms are available for the rapid and cost effective sequencing and thereby for the comparative genome analysis of *Aeromonas* spp. (Tekedar et al., 2019). Antibiotic resistance identification databases such as the CARD has been used widely to predict the resistance genes from the genome or metagenome data and can also predict the possible SNPs. Such computational predictions can have the promises to identify the presence of resistance genes and its mode of transfer based on its location in the genome or plasmid. Genome data could also be used for mapping the virulence and also for population studies of pathogens (Kumar G et al., 2017). Hence, the genomic analysis of emerging pathogens like *A. veronii* is highly significant. At the sametime, mRNA or transcriptome sequencing could also be used to study the expression rate of virulence and antibiotic resistance factors in multidrug resistant pathogens.

In the study, previously identified *Aeromonas veronii* XhG1.2 was subjected to whole genome sequencing using illumina platform to understand the genomic basis of virulence factors and antimicrobial resistance. The obtained sequence data was further assembled and annotated using various bioinformatics tools. By this, open reading frames were identified from the assembly prior to the prediction of virulence genes and antimicrobial resistance genes (Pang et al., 2015). The results of the study indicate the significance of genome based analysis to predict the environmental threat from emerging pathogens of genus *Aeromonas*.

2 MATERIALS AND METHODS

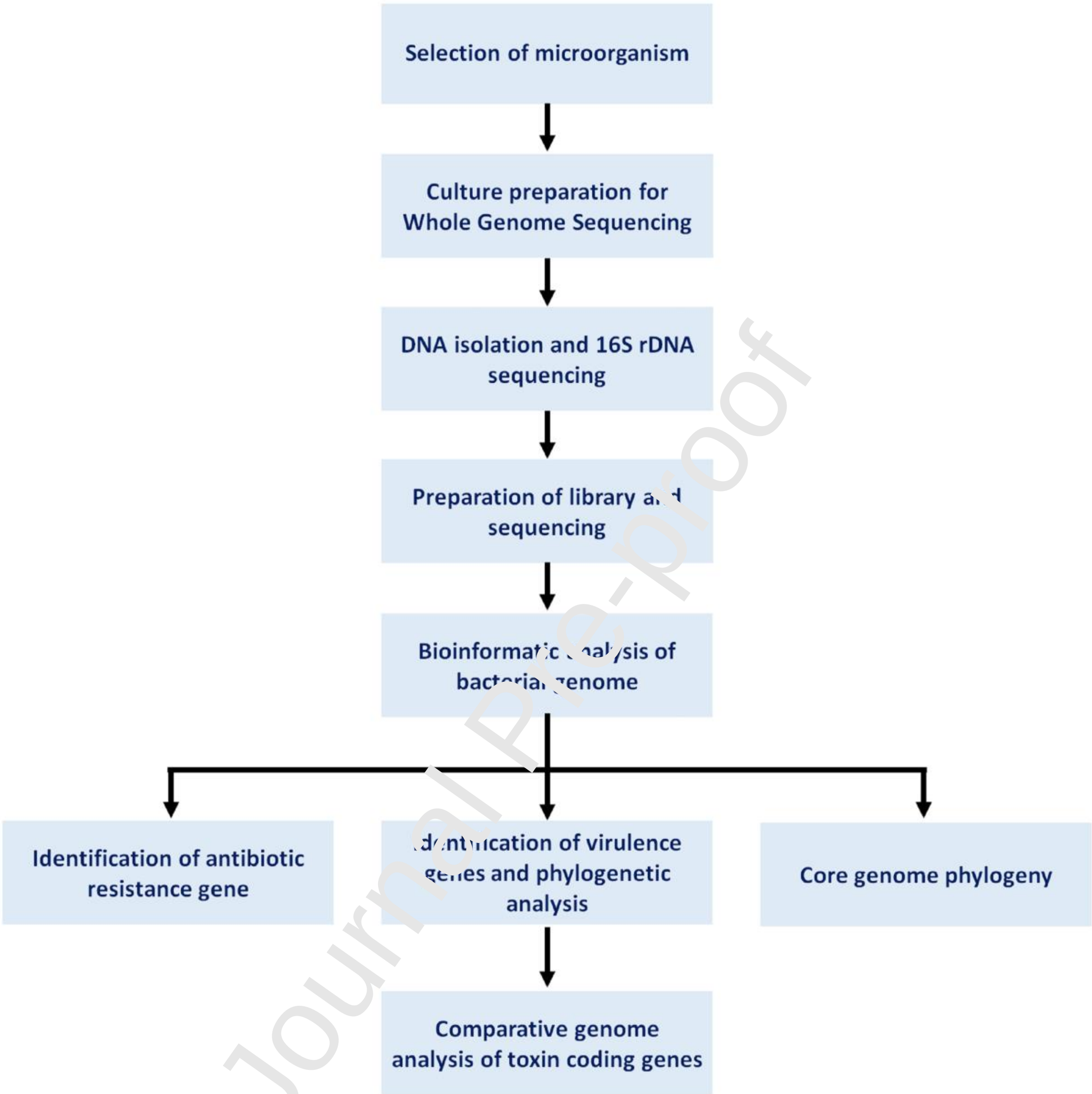

Fig. 1 Schematic representation of methods used for the study

2.1 Selection of microorganism

Aeromonas veronii XhG1.2 isolated and characterized from diseased *Xiphophorus hellerii* was selected for the whole genome sequencing. The pathogenic potential and characterization of the selected bacterium has already been described in our previous study (Das et al., 2020). As the pathogen was demonstrated to have multiple virulence factors and caused disease in fish within a period of 24 h, it was considered to be hyper-virulent.

2.2 Culture preparation for Whole Genome Sequencing

For this, XhG1.2 was inoculated into 1 mL of Luria Bertani broth (Himedia-Cat. No. M1245) and incubated overnight at room temperature. The culture was then centrifuged at 10000 rpm for 10 min to pellet the cells. The collected cells were then washed and re-suspended in sterile 1X phosphate buffered saline (PBS) and its optical density was adjusted to 1 at 600nm. From this, 1 mL aliquots were collected in micro centrifuge tube and used for further studies (Busch, Tomaso, & Methner, 2019).

2.3 DNA isolation and 16S rDNA sequencing

For the WGS analysis, genomic DNA (gDNA) was isolated from the *A. veronii* XhG1.2 using Xcelgen Bacterial gDNA Isolation kit (Xcelris- Cat No: XG2411-01) as per the manufactures instruction. Quality of isolated gDNA was checked on 0.8% agarose gel. From the isolated gDNA, 1 µL was used for determining the DNA concentration using Qubit® 2.0 Fluorometer (Khanna & Mohanty, 2017). The 16S rDNA was further amplified from the DNA using the primers 27 F (5'- AGAGTTTGGATCMTGGCTCAG–3') and 1492R (5'-CGGTTACCTTGTTACGACTT -3'). The PCR conditions were initial denaturation at 95 °C for 5 min; 35 cycles of denaturation (30 sec at 94 °C), annealing (30 sec at 55 °C), and extension (2 min at 72 °C); and a final extension at 72 °C for 7 min (Jiang et al., 2006). The PCR amplicon was purified and further processed for sequencing. Forward and reverse DNA sequencing reactions were carried out with 27F and 1492R primers using BDT v3.1 Cycle sequencing kit on ABI 3730xl Genetic Analyzer (Dixit, Appu Kuttan, & Shrivastava, 2018).

2.4 Preparation of library and sequencing

The paired-end sequencing library was prepared using *Truseq Nano DNA Library prep kit* (Illumina- Cat. No. FC-121-4001) by following the manufacturers instruction. The library preparation process was initiated with 200 ng DNA and the DNA was mechanically sheared into smaller fragments. This was followed by the end-repair by adding 'A' to the 3' ends to make the DNA fragments ready for adapter ligation at both ends. The adapter ligated DNA fragments with specific adapters were then amplified by polymerase chain reaction using HiFi PCR Master Mix using the standard illumina primers. The library was sequenced on Illumina platform (2 x 150 bp chemistry) to generate ~1.5 GB data (Chaudhry & Patil, 2016). All these were carried out at the NGS service facility of Xcelris Labs Ltd. Ahmedabad, Gujarat.

2.5 Bioinformatic analysis of bacterial genome

De novo assembly of high quality PE reads of *A. veronii* XhG1.2 genome was accomplished using the Velvet v1.2.10 (k-mer 99) (https://www.ebi.ac.uk/~zerbino/velvet/). The scaffolds were further gap filled using PE reads information by GapCloser v-1.12 software (https://www.westgrid.ca/support/software/gapcloser)(Zerbino & Birney, 2008). The protein coding genes were predicted from the XhG1.2 genome using the GenemarkS ORF prediction program (http://exon.gatech.edu/GeneMark/genemarks.cgi), a self-training program using Hidden Markov Model. The final assembled genome was given as input and executed in prokaryotic mode (Besemer, 2001).The predicted 4,165 ORFs were subjected to similarity search against NCBI's non-redundant (NR) database using the BLASTx algorithm with an E-value of 1e-05 (Chaudhry & Patil, 2016). CIRCOS 0.67-4 (http://circos.ca/) was used to generate the circular map and visual overview of the Scaffolds, ORFs, COG, tRNAs, rRNA and GC% of the draft genome of XhG1.2 (Koch & Wiese, 2012).

2.6 Identification of virulence genes and phylogenetic analysis

Identification of virulence genes was carried out using the BLASTx (e-value threshold of 1e-05) search against the protein sequences of virulence factor database (VFDB-http://www.mgc.ac.cn/VFs/main.htm) (Chen et al., 2005). The phylogenetic tree of virulence genes present in XhG1.2 such as aerolysin, hemolysin III and thermo stable hemolysin was prepared using maximum likelihood method with 1000 bootstrap with MEGA X by comparing the virulence factors of *Aeromonas hydrophila*, *Aeromonas salmonicida*, and *Aeromonas veronii*.

2.7 Identification of antibiotic resistance gene

Identification of antibiotic resistant gene was carried out using the resistance gene identifier (RGI) search of The Comprehensive Antibiotic Resistance Database (CARD-https://card.mcmaster.ca/) using the protein homology model. Perfect and strict hits with an e value $<10^{-10}$ were considered for the analysis (Tekedar et al., 2019).

2.8 Comparative Genomic analysis of toxin coding genes

For the determination of toxin coding genes in different *Aeromonas* species, *A. veronii* XhG1.2 and other 29 complete genome sequences including *A. hydrophila* (10), *A. veronii* (9) and *A. salmonicida* (10) were selected from NCBI (**Supplementary Table 1)**. The toxin encoding genes of XhG1.2 were also obtained from the virulence factor database and compared against the database comprised of 29 complete genome sequences using the Blast2 program.

2.9 Core genome phylogeny

SNP based core genome phylogeny was used to give more resolution to the phylogeny of pathogens selected from various sources. For constructing the SNP based core genome phylogeny of XhG 1.2, 29 complete genome sequences of *A. hydrophila* (10), *A. veronii* (9) and *A. salmonicida* (10) were obtained from the National Centre for Biotechnology Information (NCBI) database. The genomes were aligned using muscle program, and a core-genome phylogenetic tree was generated by Parsnp (https://github.com/marbl/parsnp) in the Harvest package (Treangen et al., 2014). The parsnp output was visualised by interactive platform Gingr (http://github.com/marbl/gingr) and analysed the SNP based core genome phylogeny.

3 RESULTS

3.1 DNA isolation and 16S rDNA sequencing

The gDNA isolated from *A. veronii* XhG1.2 was analysed by agarose gel electrophoresis which showed the presence of good quality DNA (**Supplementary Fig. 1**). Further quantitative analysis showed the concentration of DNA purified to be 28 ng/µL. The 16S rDNA amplification of XhG1.2 was resulted in the formation of amplicon of 1500bp size. The NCBI-BLAST analysis of the 16S rDNA sequence of XhG1.2 further confirmed its highest identity to *Aeromonas veronii.*

3.2 Preparation of library and Sequencing

The library of XhG1.2 was prepared and the average size of the library was found to be 426 bp**.** Sequencing of the prepared library on illumina platform has resulted in the generation of raw reads. The read statistics of data showed 15,682,190 paired ends reads with a total base length of 2333967213 or 2.3 GB data size.

3.3 Bioinformatic Analysis

3.3.1 *De novo* Assembly and ORF prediction

De novo assembly of high quality PE reads of XhG1.2 showed a total number of 34 scaffolds with a total scaffold length of 4573855 bp. The G+C content was also calculated to be 58.7%. The protein coding gene prediction of WGS data of XhG1.2 showed a total of 4165 genes as identified using the GenemarkS program. The mean ORF size was 946, and also 4110 genes were found to have hit with the reference database. The total ORF length was 3,941,505 base pairs and maximum ORFs length was 10,689 base pairs. Circular representation of the genome was constructed using the CIRCOS 0.67-4 (**Supplementary Fig. 2, 3 and 4**) and the genome was submitted to NCBI with the accession no. PRJNA650139.

3.3.2 Functional annotation of predicted ORFs

The predicted 4,165 ORFs were subjected to similarity search against the NCBI's non-redundant (NR) database. The statistics of BLAST annotation of XhG1.2 and the top-hit species distribution revealed majority of the hits to be against the species *Aeromonas veronii*

3.3.4 Identification of virulence genes of XhG1.2 and phylogenetic analysis of virulence genes.

BLASTx analysis of XhG1.2 WGS data with the protein sequences of virulence factor database (VFDB) resulted in the identification of 1100 virulence genes. Out of 1100 genes, 56 genes were 100 % similar to the VFDB and 77 genes showed similarity in the 90-99% range. The virulence genes such as aerolysin, haemolysin III and thermo stable haemolysin of the XhG1.2 were compared with other reported genes and used for phylogenetic analysis (**Fig. 2)**.

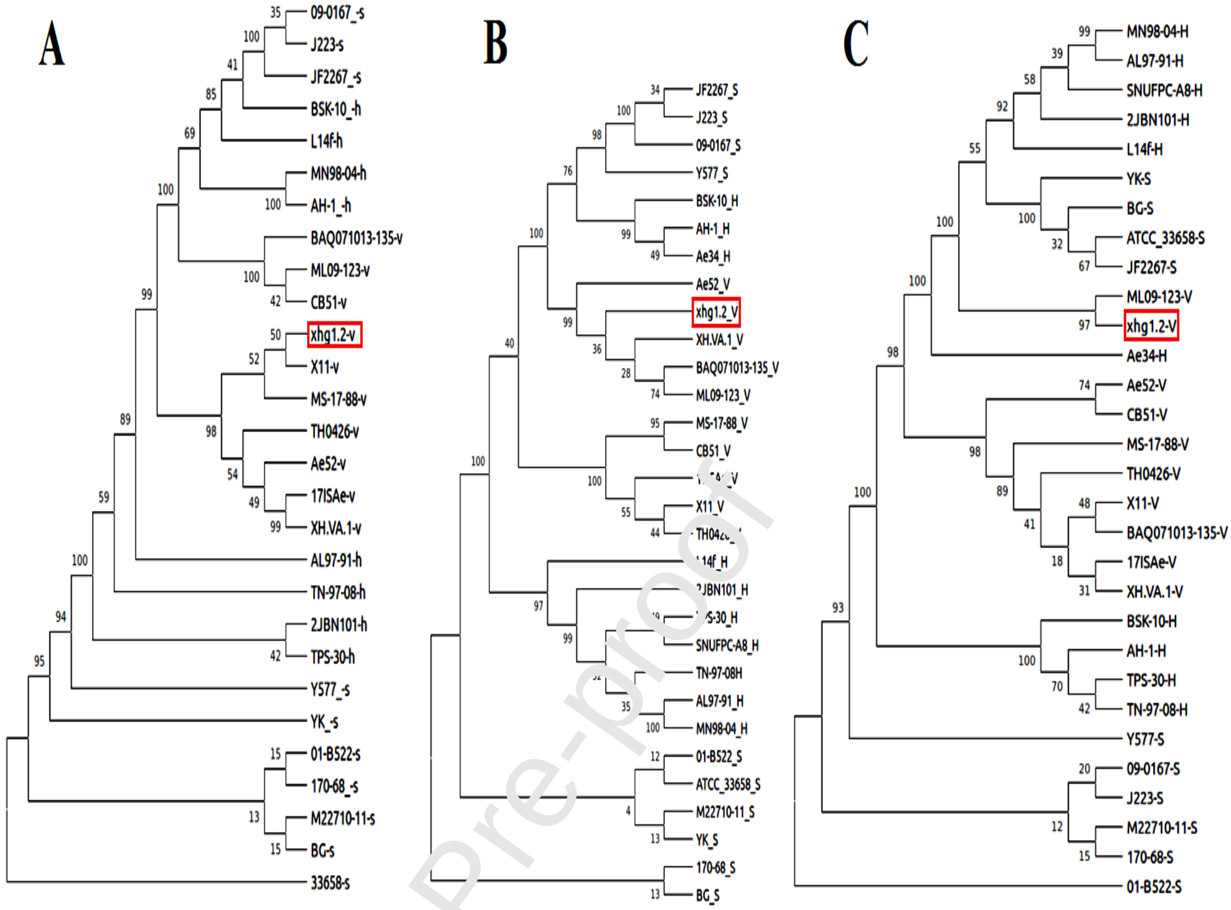

Fig. 2 Phylogenetic-analysis of virulence factors of XhG1.2 along with sequence from databases using Maximum Likelihood method (MEGA X). A, aerolysin; B, hemolysin III and C, thermostable haemolysin.

3.3.5 Antibiotic resistance gene identification

The BLAST analysis in CARD reference data has resulted in the identification of 4 different genes coding for antibiotic resistance. These were CphA3 for carbapenem resistance, OXA-12 for cephalosporins resistance, *Aed*F for fluoroquinolone and tetracycline resistance and also the gene for Pulvomycin resistance (**Fig. 3 and Table 1**).

Fig. 3 RGI circular plot showing the antimicrobial resistance genes identified from *A. veronii* XhG1.2. The identified genes show strict hits against 4 different resistance coding genes.

Table 1 Antibiotic resistance genes identified from WGS data of XhG1.2 using CARD RGI finder

RGI criteria	ARO Term	Detection Criteria	AMR Gene Family	Drug Class	Resistance Mechanism	% Identity
Strict	cphA3	protein homolog model	CphA beta-lactamase	carbapenem	antibiotic inactivation	99.21
Strict	OXA-12	protein homolog model	OXA beta-lactamase	cephalosporin, penam	antibiotic inactivation	96.96
Strict	Pulvomycin resistance	protein variant model	elfamycin resistant EF-Tu	elfamycin antibiotic	antibiotic target alteration	90.84
Strict	Pulvomycin resistance	protein variant model	elfamycin resistant EF-Tu	elfamycin antibiotic	antibiotic target alteration	89.06
Strict	adeF	protein homolog model	RND antibiotic efflux pump	fluoroquinolone antibiotic, tetracycline antibiotic	antibiotic efflux	48.65
Strict	adeF	protein homolog model	RND antibiotic efflux pump	fluoroquinolone antibiotic, tetracycline	antibiotic efflux	43.8

				antibiotic		

3.3.6 Comparative genomic analysis

The comparative genome analysis revealed the presence and distribution of toxin coding genes in *A. veronii* XhG1.2. The percentage identity of the genes showed the genome of *A. veronii* XhG1.2 to have differences in gene sequences when compared with *A. hydrophilla* and *A. salmonicida* **(Fig. 4)**. A heat map was constructed for the visualisation of percentage identity of the genes in colour gradients.

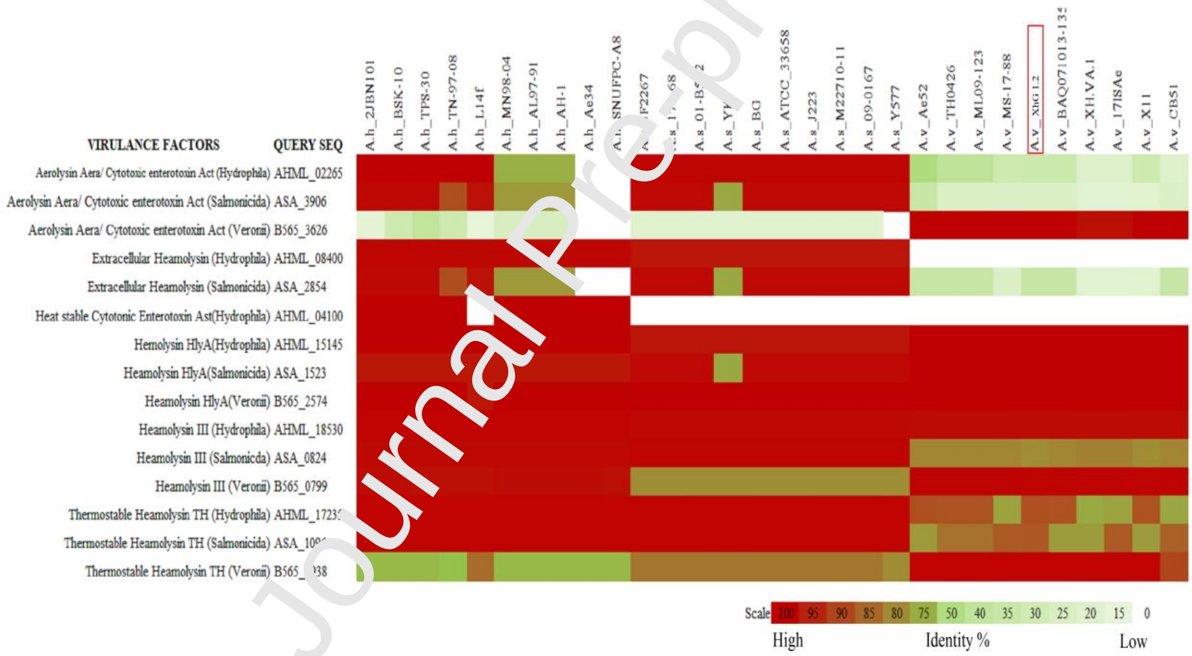

Fig. 4 Heat map representation of comparative genome data of *Aeromonas veronii* against *A. hydrophilla*, and *A. salmonicida*. *A. veronii* XhG 1.2 genome is marked under the red box. The scale shows red to white colour gradient for 100 to 0% identity respectively.

3.3.7 Core genome phylogeny

The phylogenetic analysis of XhG1.2 with 29 other *Aeromonas* genomes was done based on core genome SNP prediction. The phylogenetic tree clearly distinguished the three different species of *Aeromonas* spp. in which *A. veronii* XhG1.2 was clustered with *A. veronii* X11 (**Fig.5)**.

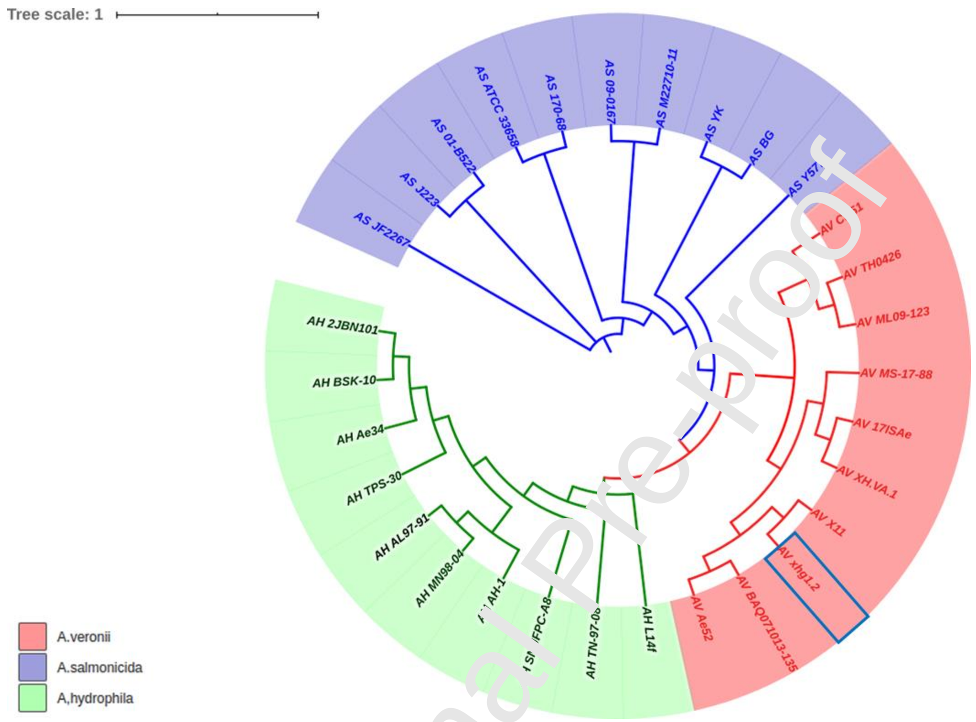

Fig.5 The SNP based phylogenetic analysis of 30 *Aeromonas* genomes including members of *A. veronii, A. salmonicida* and *A. hydrophilla* with XhG1.2.

4 DISCUSSION

Dynamic nature of the aquatic environment makes it highly challenging to the microorganisms. This makes them to have the capability to develop adaptive strategies. In the case of fish pathogens like *Aeromonas* spp., these adaptations result in the evolution of multi-virulence and antibiotic resistance properties. Eventhough such mechanisms are well described for *A. hydrophila*, the same for other *Aeromonas* spp. is just began to explore.

Aeromonas veronii is an emerging pathogen which has recently been identified to cause variety of fish diseases like ulcerative syndrome and haemorrhagic infections. In the present study, the whole genome sequence data analysis of *Aeromonas veronii* XhG1.2 has been carried out for the identification of genetic basis of virulence and antibiotic resistance. This is because, the *A. veronii* XhG1.2 has already been demonstrated to have hyper-virulent nature (Das et al., 2020). As the genome of *A. veronii* is expected to codes for different toxins such as aerolysin, RTxA and secretory systems such as T2SS and T3SS, a detailed genomic analysis is essential to predict its threat to aquaculture.

Members of *Aeromonas* spp. are already reported to have diverse virulence factors. The aerolysin present in them can attack the intestinal epithelium of fishes and could make pores on the lipid bilayer. Bacterial entry to the circulatory system through this can ultimately result in septicemia (Ran et al., 2018). Aerolysin has also been described in *A. hydrophila*, *A. salmonicida*, *A. caviae* and other members of Aeromonadales (Bhowmick & Bhattacharjee, 2018). Likewise, the *Aeromonas veronii* XhG1.2 has also been described to have the distribution of large number of virulence factors with high risk to the health of wide range of ornamental fishes. This demands the need for detailed genomic investigation of XhG1.2 through WGS analysis.

All the major conserved secretory systems were found to be present in XhG1.2 genome. These include the T2SS, T3SS, and T6SS which are also reported to be present in most of the *Aeromonas veronii* genome. The secretory systems are considered to function by secreting the virulence factors such as toxins and hydrolytic enzymes (Korotkov & Sandkvist, 2019). As the XhG1.2 has already been observed to have various virulence factors, the observed secretory systems might also play important role in the diseases caused by the bacteria. This is because XhG1.2 was originally isolated from diseased *Xiphophorus hellerii*.

T2SS system is present mainly in Gram negative bacteria and it functions by secreting toxins and enzyme (Korotkov & Sandkvist, 2019). However, T3SS forms channel like pili connecting the bacteria with host cytoplasmic membrane. This enables the injection of the toxins and the partially folded proteins directly into the host cell (Deng et al., 2017).

Furthermore, the pathogenicity of bacteria can also be influenced by the presence of structures such as flagella, fimbriae and pili. These help the pathogen to adhere and invade the host rapidly. The XhG1.2 genome also showed the evidence for the presence of lateral (gene 1705) and polar flagella (gene 1799) which are considered to be involved in swimming and swarming motility. The total biosynthetic genes for lateral and polar flagella were 46 and 121 respectively. For the adherence of bacterial cells on to the host surface, there are type IV pili groups such as Flp type IV pili, MSHA type IV pili and Tap type IV pili. The flagella and pili can also help in the biofilim formation which is an important pathogenicity determinant (Kirov, Castrisios, & Shaw, 2004). The motility structures of XhG1.2 can also consider to play significant role in its ability to cause diseases in ornamental fishes as it favor the attachment of bacteria onto the fish to initiate the infection.

The antibiotic resistance in *A. veronii* XhG1.2 was also analysed using the resistance gene identifier (RGI) of the comprehensive antibiotic resistance database (CARD). The major genes identified from the RGI were those which code for the resistance against multiple antibiotics such as carbapenems, cephalosporins, fluoroquinolones and tetracycline. The *cph*A3 and OXA 12 predicted from the XhG1.2 genome code for resistance against carbapenems and cephalosporins respectively. OXA-12 gene is common among *Aeromonas* species and has also been reported in *A. jandaei* (Alksne & Rasmussen, 1997). The function of OXA-12 involves the inactivation of cephalosporin antibiotics which are often used in aquaculture treatments. *Cph*A3 or *Cph*A Beta-lactamase gene of XhG1.2 codes for the proteins involved in the inactivation of the carbapenem class of antibiotics (Villadares et al., 1996). Carbapenems such as ertapenem, meropenem and imipenem are the important drugs in this class, which are used against Gram-negative bacteria. These drugs generally function by inhibiting the cell wall synthesis.

There were two copies of genes coding for the elfamycin resistance (Pulvomycin resistance gene) with the similarity of 89.06% and 90.84% to CARD database (ARO :3003369) (Zeef et al., 1994). This resistance is coded by the Pulvomycin resistance gene which has previously been reported in *E. coli*. Elfamycins are the less studied antibiotics which act by inhibiting the process of translation. The mechanism of resistance for the same generally involves the alteration of antibiotic target (Prezioso, Brown, & Goldberg, 2017). The XhG1.2 genome has also shown to have the genes coding for resistance to fluoroquinolone and tetracycline antibiotics due to the presence of *aed*F gene which is a Resistance Nodulation and cell Division (RND) antibiotic flux pump. RND family proteins are group of efflux pump in Gram-negative bacteria, which could potentially code for heavy metal and antibiotic resistance. The same gene has already been reported in *Acinetobacter baumannii* with multidrug resistance (Asif, Alvi, & Ur Rehman, 2018). The observed resistance to various antibiotics indicates the possible challenges with treatment of diseases caused by *A. veronii* XhG1.2. The presence of these antibiotic resistance genes could either contribute to the complete resistance or may increase the minimum inhibitory concentrations (Friedman, Alder, & Silverman, 2006; Vila et al., 1994). This also indicates the challenge with the transfer of this resistance from XhG1.2 to other microorganisms including those which cause potential risk to humans.

The WGS data supports the pathogenicity of XhG1.2 isolated in the previous study by confirming the presence of wide range of virulence factors associated with toxin production, secretion, motility and adherence. The comparative analysis of *Aeromonas* genomes shows the uniqueness of the *A. veronii* genome when compared to *A. hydrophila* and A. *salmonicida*. Aerolysin gene sequence of XhG1.2 was more similar to those from other *A. veronii* than with *A. hydrophila* and *A. salmonicida*. Hemolysin III and thermostable hemolysin of *A. veronii* showed only 80 % to 90 % similarity with *A. hydrophila* and *A. salmonicida.* The phylogenetic analysis also supports the evolutionary changes of *A. veronii* genomes. This indicates the XhG1.2 to be highly pathogenic and is further challenged by the resistance to several antibiotics. The acquisition of the observed resistance genes in XhG1.2 might be due to the unscientific usage of antibiotics in aquaculture or through homologous gene transfer from other multidrug resistant pathogens. However, the antibiotic usage in the aquaculture could potentially enhance the resistance emergence and hence management of emerging multi-virulent organisms like *A. veronii* XhG1.2 isolated in the study will be highly challenging.

5 Conclusions

The whole genome sequencing of the *A. veronii* XhG1.2 was conducted and the genes were annotated. The genome analysis of XhG1.2 revealed the presence of potential virulence factors coding for toxins such as aerolysin, RtxA, secretory systems, motility and adherence structures. The genome also showed the presence of resistance genes coding for resistance against carbapenem, cephalosporin and fluoroquinolone. The results confirm the pathogenic potential of XhG1.2 and its features as a successful fish pathogen.

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Supplementary data

Supplementary material

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Highlights

- The evolution of antibiotic resistance in aquatic bacteria *Aeromonas veronii* XhG1.2
- Genomic DNA sequencing and 16S rDNA analysis
- COG analysis showed 304 general function genes &285 amino acid transport and metabolism
- genes
- Major virulence factors predicted as aerolysin, RtxA, T2SS, T3SS, T6SS, adherence and motility
- major antibiotic resistance genes found are *Cph*A3, OXA-12, *aed*F and Pulvomycin